INCH-POUND
MIL-DTL-32139/9A
w/AMENDMENT 1
21 March 2022
SUPERSEDING
MIL-DTL-32139/9A
11 February 2020

DETAIL SPECIFICATION SHEET

CONNECTORS, ELECTRICAL, RECTANGULAR, NANOMINIATURE, DUAL ROW, PLUG, REAR PANEL MOUNT, POLARIZED SHELL, PIG TAIL, PIN CONTACTS, CRIMP TYPE, INSULATED WIRE, GASKET SEAL

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-32139.

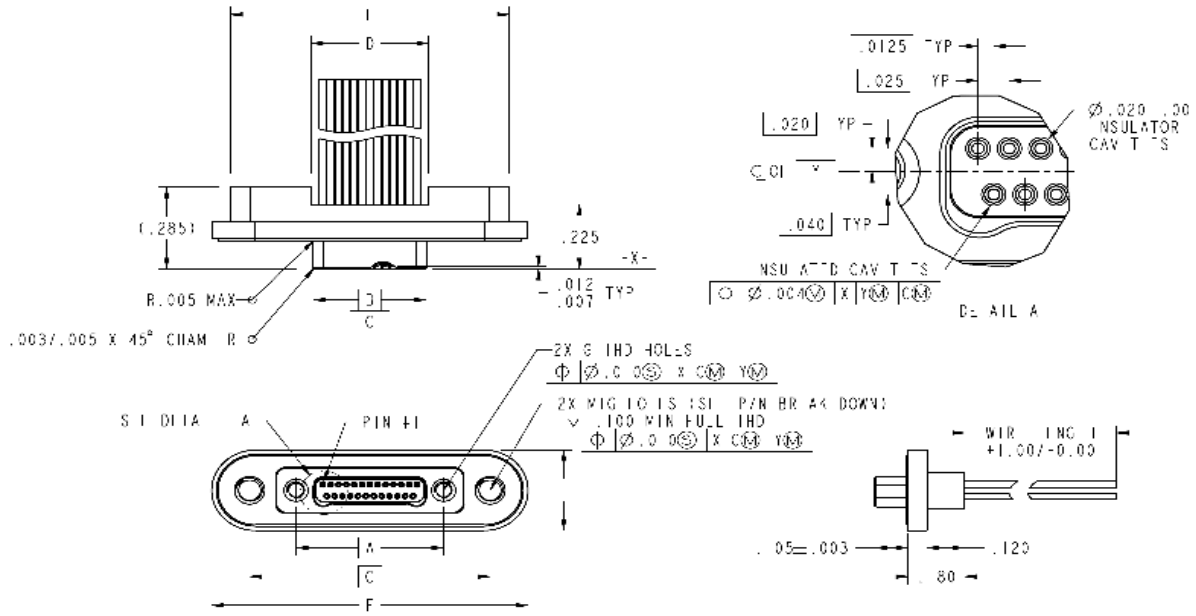
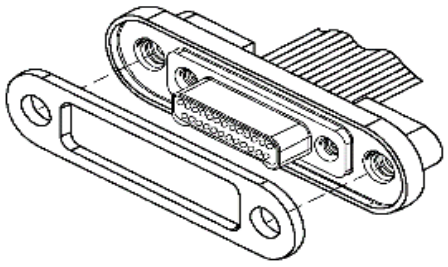


FIGURE 1. Plug, Rear Panel Mount dimensions and configurations.



Shown with Gasket

AMSC N/A
FSC 5935

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Size	A BSC	B BSC	C BSC	D	E	F	G	H
9	.270 (6.86)	.160 (4.06)	.566 (14.38)	.175 (4.45)	.688 (17.48)	.808 (20.52)	#0-80 UNF-2B	.280 (7.11)
15	.345 (8.76)	.235 (5.97)	.641 (16.28)	.250 (6.35)	.763 (19.38)	.883 (22.43)	#0-80 UNF-2B	.280 (7.11)
21	.420 (10.67)	.310 (7.87)	.716 (18.19)	.325 (8.26)	.838 (21.29)	.958 (24.33)	#0-80 UNF-2B	.280 (7.11)
25	.470 (11.94)	.360 (9.14)	.766 (19.46)	.375 (9.53)	.888 (22.56)	1.008 (25.60)	#0-80 UNF-2B	.280 (7.11)
31	.545 (13.84)	.435 (11.05)	.841 (21.36)	.450 (11.43)	.963 (24.46)	1.083 (27.51)	#0-80 UNF-2B	.280 (7.11)
37	.620 (15.75)	.510 (12.95)	.916 (23.27)	.525 (13.34)	1.038 (26.37)	1.158 (29.41)	#0-80 UNF-2B	.280 (7.11)
51	.795 (20.19)	.685 (17.40)	1.091 (27.71)	.700 (17.78)	1.213 (30.81)	1.333 (33.86)	#0-80 UNF-2B	.280 (7.11)
65	.970 (24.64)	.860 (21.84)	1.266 (32.16)	.875 (22.23)	1.388 (35.26)	1.508 (38.30)	#0-80 UNF-2B	.280 (7.11)
69	1.020 (25.91)	.910 (23.11)	1.316 (33.43)	.925 (23.50)	1.438 (36.53)	1.558 (39.57)	#0-80 UNF-2B	.280 (7.11)
85	1.246 (31.65)	1.110 (28.19)	1.568 (39.83)	1.125 (28.58)	1.690 (42.93)	1.810 (45.97)	#2-56 UNC-2B	.305 (7.75)

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Unless otherwise specified tolerances are ±.005 inch (0.13 mm) angular tolerance ±2°.
4. Shell shall be flush to insulator within ±.004 inch (0.10 mm).
5. 30 AWG wire is the largest wire size that can be used in the connector assembly

FIGURE 1. Plug, Rear Panel Mount dimensions and configurations - Continued.

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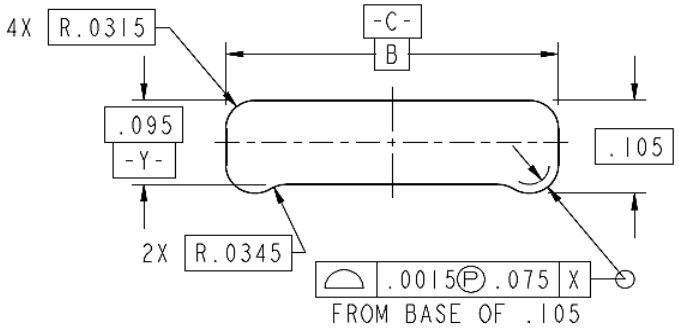
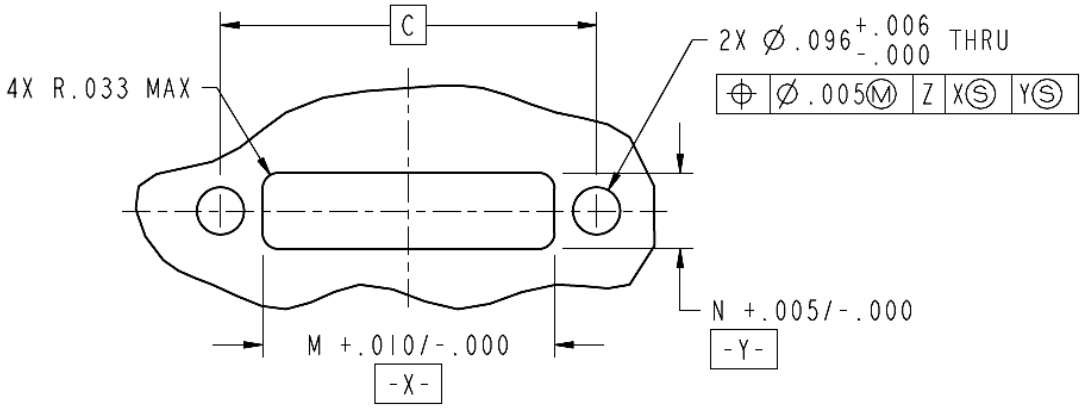


FIGURE 2. Nano Connector Interface.



Size	C BSC	M	N
9	.566 (14.38)	.395 (10.03)	.155 (3.94)
15	.641 (16.28)	.470 (11.94)	.155 (3.94)
21	.716 (18.19)	.545 (13.84)	.155 (3.94)
25	.766 (19.46)	.595 (15.11)	.155 (3.94)
31	.841 (21.36)	.670 (17.02)	.155 (3.94)
37	.916 (23.27)	.745 (18.92)	.155 (3.94)
51	1.091 (27.71)	.920 (23.37)	.155 (3.94)
65	1.266 (32.16)	1.095 (27.81)	.155 (3.94)
69	1.316 (33.43)	1.145 (29.08)	.155 (3.94)
85	1.568 (39.83)	1.397 (35.48)	.180 (4.57)

FIGURE 3. Panel cutout dimensions and configurations.

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NOTES:
1. Panel cutout is sized to allow connector mounting with either lobes up or down orientation.
2. Recommended panel thickness .100 MAX.

FIGURE 3. Panel cutout dimensions and configurations - Continued.

REQUIREMENTS:

Dimensions and configuration see figures 1, 2, and 3.

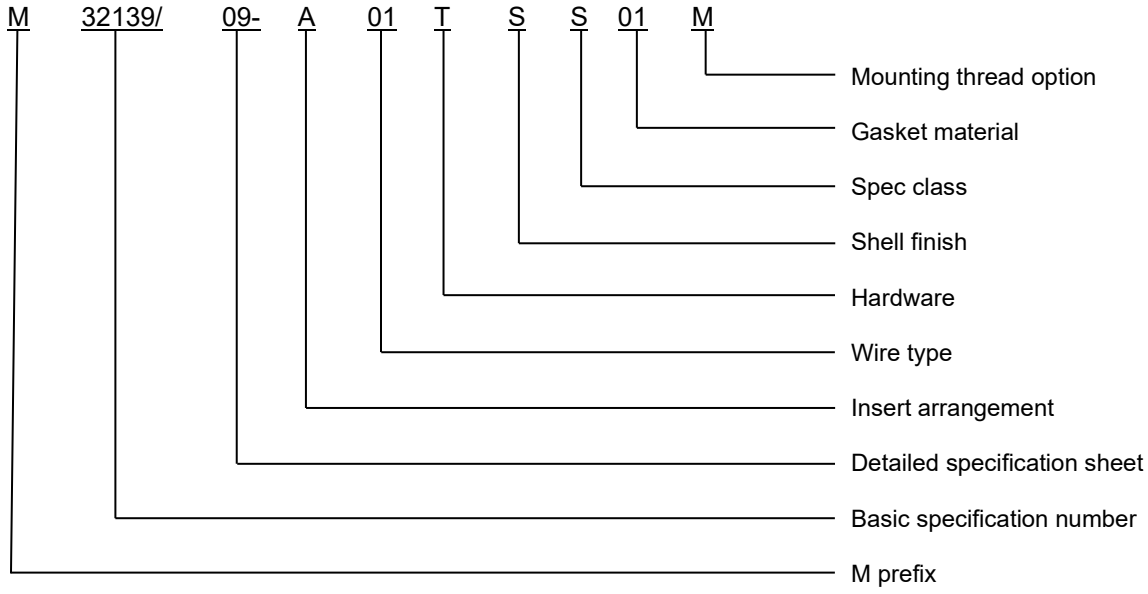
This specification sheet describes the pin side of a rectangular connector. This connector uses reverse gender contacts, i.e., the live pin recessed in the insulator with the static socket protruding from a shrouded interface.

Contact connection: The pin contact, which is recessed in the insulator, is normally connected to the live side of the circuit.

To protect the electrical enclosure equipment, the panel mount connector shall seal from the outside environmental conditions such as dust and splashed water.

Interface dimensions according to MIL-DTL-32139/3.
Mating receptacle: Shall be in accordance with MIL-DTL-32139/4.

Part of Identifying Number (PIN):



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Insert arrangement	Wire type [1/]	
A = 9	01 = NEMA HP3-ETXBBB9	6 inches (152 mm) long
B = 15	02 = NEMA HP3-ETXBBB9	18 inches (457 mm) long
C = 21	03 = NEMA HP3-ETXBBB9	36 inches (914 mm) long
D = 25	04 = NEMA HP3-ETXBBB( )	6 inches (152 mm) long [2/]
E = 31	05 = NEMA HP3-ETXBBB( )	18 inches (457 mm) long [2/]
F = 37	06 = NEMA HP3-ETXBBB( )	36 inches (914 mm) long [2/]
G = 51	07 = M22759/33-30-9	6 inches (152 mm) long [3/ 4/]
H = 65	08 = M22759/33-30-9	18 inches (457 mm) long [3/ 4/]
J = 69	09 = M22759/33-30-9	36 inches (914 mm) long [3/ 4/]
K = 85	10 = M22759/33-30-( )	6 inches (152 mm) long [2/ 3/ 4/]
	11 = M22759/33-30-( )	18 inches (457 mm) long [2/ 3/ 4/]
	12 = M22759/33-30-( )	36 inches (914 mm) long [2/ 3/ 4/]
	13 = 04047-30A-9	6 inches (152 mm) long
	14 = 04047-30A-9	18 inches (457 mm) long
	15 = 04047-30A-9	36 inches (914 mm) long
	16 = 04047-30A-( )	6 inches (152 mm) long [2/]
	17 = 04047-30A-( )	18 inches (457 mm) long [2/]
	18 = 04047-30A-( )	36 inches (914 mm) long [2/]
	19 = M22759/47-30-9	6 inches (152 mm) long
	20 = M22759/47-30-9	18 inches (457 mm) long
	21 = M22759/47-30-9	36 inches (914 mm) long
	22 = M22759/47-30-( )	6 inches (152 mm) long [2/]
	23 = M22759/47-30-( )	18 inches (457 mm) long [2/]
	24 = M22759/47-30-( )	36 inches (914 mm) long [2/]
	25 = M22759/51-30-9	6 inches (152 mm) long
	26 = M22759/51-30-9	18 inches (457 mm) long
	27 = M22759/51-30-9	36 inches (914 mm) long
	28 = M22759/51-30-( )	6 inches (152 mm) long [2/]
	29 = M22759/51-30-( )	18 inches (457 mm) long [2/]
	30 = M22759/51-30-( )	36 inches (914 mm) long [2/]
	31 = M22759/52-30-9	6 inches (152 mm) long
	32 = M22759/52-30-9	18 inches (457 mm) long
	33 = M22759/52-30-9	36 inches (914 mm) long
	34 = M22759/52-30-( )	6 inches (152 mm) long [2/]
	35 = M22759/52-30-( )	18 inches (457 mm) long [2/]
	36 = M22759/52-30-( )	36 inches (914 mm) long [2/]

Hardware	Shell and finish	Space class
T = Female Thread	S = Passivated stainless steel shell T = Titanium	Blank for non-space applications. S = Space class Mounting Thread Option Omit For #2-56 UNC-2B M = M2X0.4 6H

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1/ Pig tail wire lead tolerance is +1.00 inch/-0.0 inch (25.4/-0.0 mm).
2/ Color coding in accordance with MIL-STD-681, system 1, except using ten solid colors only in repeating sequence.
3/ Corrosion has been experienced on M32139 connectors that are pre-wired with M22759/33-30-9 or M22759/33-30-() wires and stored in a sealed environment. CAUTION SHOULD BE EXERCISED WHEN USING THIS WIRE.
4/ Wire M22759/33 use insert wire strip tool SAE AS5768/1, T7 or B7 for 30 gauge wire.

Gasket Material
Omit for no gasket
01 = Fluoroslicone IAW MIL-DTL-25988, Type II, Class I, Grade 70
02 = Passivated Silver Plated Aluminum Filled Fluorsolicone IAW MIL-DTL-83528, Type “D” (CHO-Seal 1298 or Equivalent)
03 = Nickel Plated Aluminum Filled Fluorosilicone, (CHO-Seal 6503 or Equivalent)

Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Referenced documents. In addition to MIL-DTL-32139, this document references the following:

MIL-DTL-32139/3
MIL-DTL-32139/4
MIL-DTL-25988
MIL-DTL-83528
MIL-STD-681
SAE AS5768/1

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air Force - 85
DLA - CC

Review activities:
Army - MI

Preparing activity:
DLA - CC

(Project 5935-2022-064)

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